

Equal exponents give one logarithm for every (k, h) :

$$Z(n) \sim \frac{A_{p-1}}{2k_1 k_2} n^{-p/2} \log n$$

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Abstract

Unit 44 reduced the general two-dimensional chart integral exactly to $\frac{1}{k_1 k_2} n^{-p_1/2} b^{k_2(p_2-p_1)} \int_0^L x^{p_1-p_2-1} F_{p_2-1}$. This unit supplies the analytic infrastructure for the weighted primitives F_γ ($\gamma > -1$) and settles the equal-exponent regime $p_1 = p_2 = p$: with $A_\gamma = \int_0^\infty t^\gamma f = S_{(\gamma+1)/2}(a)/2$,

$$Z(n) \sim \frac{A_{p-1}}{2k_1 k_2} n^{-p/2} \log n.$$

At $k = (1, 1)$, $h = (0, 0)$ this is unit 35. Verified numerically at $k = (2, 2)$, $h = (2, 2)$ (ratio to the leading term $\rightarrow 1$). Zero `sorry/axiom`.

1 Setting

Everything in unit 35 used three facts about $F = F_0$: it is monotone with $F \leq \min(A, Ex)$, its tail is $A - F(x) \leq M/x$, and hence $\int_1^L F/x = A \log L + O(1)$. The weighted analogues are $F_\gamma \leq \min(A_\gamma, Ex^{\gamma+1}/(\gamma+1))$ and $A_\gamma - F_\gamma(x) \leq A_{\gamma+1}/x$ (from $t^\gamma \leq t^{\gamma+1}/x$ on $t \geq x$), giving $A_\gamma \log L - A_{\gamma+1} \leq \int_0^L F_\gamma/x \leq A_\gamma \log L + E/(\gamma+1)^2$.

2 The things formalised

```

noncomputable def weightedMass ( a : ℝ ) : ℝ := λ t in Ioi 0, t ^ a * quadKernel a t
theorem weightedMass_eq ( a : ℝ ) :
  weightedMass a = 1 / 2 * fluctuation ((a + 1) / 2) a
theorem weightedKernel_integrableOn ( a : ℝ ) (h : 0 < a) (h : -1 < a) :
  IntegrableOn (fun t => t ^ a * quadKernel a t) (Ioi 0)
theorem weightedMass_sub_primitive ... (hx : 0 < x) :
  weightedMass a - weightedPrimitive a x = weightedMass a (a + 1) / x
theorem weightedLogPrimitive_bounds ( a L : ℝ ) (h : 0 < a) (h : -1 < a) (hL : 1 ≤ L) :
  weightedMass a * Real.log L - weightedMass a (a + 1)
  = weightedLogPrimitive a L
  = weightedLogPrimitive a L
  = weightedMass a * Real.log L + Real.exp ( - a ^ 2 / 2 ) / ( a + 1 ) ^ 2
theorem twoDGeneral_isEquivalent_of_eq ...
  (hp : ((h : ℝ) + 1) / k = ((h : ℝ) + 1) / k) :
  (fun n => twoDGeneral a b n h h k k) ~[atTop]
  fun n => weightedMass a (((h : ℝ) + 1) / k - 1) / (2 * ((k : ℝ) * k))
  * (n ^ (-(((h : ℝ) + 1) / k / 2)) * Real.log n)

```

3 Proof strategy

The mass identity is the substitution $s = x^2$ in the fluctuation integral via `integral_comp_rpow_Ioi_of_pos` (no hypotheses; holds even when both sides are the Bochner value of a non-integrable function), and integrability for $\gamma > -1$ comes from the same substitution through `integrableOn_Ioi_comp_rpow_iff` and the known integrability of the fluctuation integrand. Measurability of F_γ is obtained from monotonicity (`Monotone.measurable`) rather than continuity, sidestepping t^γ at negative t . The equal-exponent asymptotic follows the unit 35 template: the exact formula with $b^0 = 1$ and x^{-1} , the squeeze at $L = \sqrt{n} b^{k_1+k_2}$, $\log L = \frac{1}{2} \log n + (k_1+k_2) \log b$, and $n^r = o(n^r \log n)$ (proved once for all r from $1 = o(\log)$ via `mul_isBig0`).

4 Roadblocks and resolutions

- `intervalIntegrable_rpow'` lives in the `intervalIntegral` namespace on this pin; an unqualified call is an unknown identifier.
- `nlinarith` failed on the two-sided remainder bound because $0 \leq E/p^2$ was not in context (both are opaque atoms after `set`); adding it makes plain `linarith` succeed.
- `integral_rpow` leaves a $1 - 0$; add `sub_zero` to the rewrite chain before `field_simp`.

5 What was Mathlib, what was new

Mathlib: `integral_comp_rpow_Ioi_of_pos`, `integrableOn_Ioi_comp_rpow_iff`, `Monotone.measurable`, `setIntegral_mono_set`, `integral_rpow`, `intervalIntegral.intervalIntegrable_rpow'`, `IsLittle0.mul_isBig0`. New: the weighted mass/primitive calculus and the general equal-exponent logarithmic asymptotic.

6 Lessons

Monotonicity is the cheapest route to measurability for primitives of nonnegative integrands; it avoids every question about the integrand off the domain of interest. And when a family of lemmas (unit 35) is about to be generalised, restating them with a real parameter γ costs little more than the original once the parameter-free proofs are in hand.

7 Outcome

Two of the three regimes of the general two-dimensional reduction are now closed: equal exponents (this unit, one logarithm with exact constant) and, structurally, the all-equal d -fold case (units 39, 43). The remaining regime is $p_1 < p_2$: convergence of $\int_0^\infty x^{p_1-p_2-1} F_{p_2-1}$ and $Z(n) \sim C n^{-p_1/2}$ with C that integral, the general form of unit 37's closed form.